Outline

Rewiring the market time clock in Spanish electricity markets

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RAP

Positioning. How shorter trading intervals reshape wholesale electricity bidding and prices has been studied for Germany (Märkle-Huß, Feuerriegel and Neumann, 2017, on the 2014 EPEX 15-minute reform) and for Australia (Csereklyei and Khezr, 2024, on the 2021 five-minute settlement reform), but the recent Spanish move to 15-minute trading has not yet been evaluated. The Spanish case is distinctive: the reform was rolled out in two cutovers (intraday in March 2025, day-ahead in October 2025) and lands in the middle of a nationwide blackout (28 April 2025) and the post-blackout reinforced operation regime that has been expanding regulated redispatch since.

Research Question. How did Spain's switch from hourly to 15-minute wholesale trading change the way generators bid, how the auctions clear, and how the system as a whole settles? In particular: who actually uses the new within-hour freedom, and where does it bite? Does the reform amplify market power, or does it compress it? Does it harm consumers? And how can the day-ahead reform's effect be separated from the concurrent post-blackout regulatory regime?

Answer. The 15-minute trading reform acts not as a uniform technology shock but as a *strategic-latitude reveal*: its uptake maps to who has the economic latitude to use it strategically. Dispatchable firms (combined-cycle gas, hydro, pumped-hydro) widen their within-quarter bid ladders in high-residual-demand hours (the morning ramp and evening peak), where within-hour residual demand actually varies and rewards finer pricing; flat hours barely move. Wind and solar aggregators also use the new grid, but for a different reason — imbalance-penalty avoidance under the December 2024 settlement reform — which applies equally in every hour, and so leaves no critical-flat differential. On the auction side, intraday clearing prices drop sharply under the March 2025 reform with day-ahead co-moving down via cross-market anticipation; the day-ahead combined-cycle gas auction-cleared volume scales up by about 59 GWh/day under the October 2025 reform in critical hours; the day-ahead/intraday wedge keeps its mean at zero but its within-day standard deviation jumps in critical hours under the intraday reform. Granularity does not amplify market power (prices fall, the wedge level is unchanged) and does not harm consumers (the granularity reforms compress the real-time redispatch and reserve activations REE has to call after the auction; the post-blackout reforzada cost rise is a separate mechanism that happens to share calendar time). The interpretive frame is that the reform restructures the whole sequential-market architecture even where it does not touch the auction directly: under the intraday cutover, combined-cycle gas widens its day-ahead bid-shape *in anticipation* of finer intraday re-trading.

Questions Provoked by the RAP

1. What does the move from hourly to 15-minute trading actually change in the wholesale market, and how is the answer different for dispatchable units vs. renewables?
2. How can the day-ahead reform's effect be separated from the concurrent post-blackout reinforced-operation regime?
3. How does bid widening translate into the day-ahead combined-cycle gas cleared-volume scale-up in critical hours?
4. Does the reform amplify market power, and does it harm consumers, once both the wholesale auction and the system-cost cascade are accounted for?
5. Could the March 2025 intraday reform have contributed to the macro-conditions that made the April 2025 blackout possible?

1 Introduction

The two-step Spanish 15-minute trading reform — intraday auctions in March 2025, day-ahead market in October 2025 — arrives in an institutional setting that has been reshaped by a nationwide blackout (28 April 2025) and a continuous reinforced-operation (*reforzada*) regime that has expanded regulated redispatch ever since. We ask whether the reform changed wholesale bidding, prices and cleared quantities; whether it amplifies market power or compresses it; whether it harms consumers; and how the day-ahead cutover can be identified given the concurrent regulatory regime. The answer is that the reform is a strategic-latitude reveal: dispatchable firms widen their within-quarter bid spreads where within-hour residual demand actually varies, renewables use the finer grid uniformly to manage imbalance, the intraday auction price drops by ~ 80 EUR/MWh placebo-net under the March 2025 reform, the day-ahead combined-cycle gas cleared volume scales up by ~ 59 GWh/day under the October 2025 reform in critical hours, and a wedge-SD jump shows that within-hour bid behavior also restructures the day-ahead/intraday relationship even though the wedge's level stays at zero. The reform does not amplify market power or harm consumers: wholesale prices fall, the wedge level stays at zero, the up-direction adjustment-cost channels REE has to call shrink; the post-blackout cost rise is a separate, reforzada-driven effect that overlaps in calendar time but operates through a different mechanism (Fase I redispatch).

Questions answered by the introduction.

1. What is the puzzle, given the German and Australian precedents and the recent Spanish reform?
2. What is the paper's precise answer, and what unifying mechanism does it propose?
3. How does the rest of the paper establish the result, and what does the institutional setting force us to control for?

2 Institutional Setting: Sequential Markets, Three Reforms, and a Blackout

Spanish wholesale clearing is sequential: a day-ahead auction first (OMIE), then three intraday auctions, then continuous trading (XBID/SIDC) and balancing. The 15-minute trading reform was rolled out in three steps that interact in sequence: a 15-minute settlement reform in December 2024 (ISP15), the intraday cutover in March 2025 (ID15), and the day-ahead cutover in October 2025 (DA15). The blackout of 28 April 2025 switches the system into a continuous reinforced-operation regime that overlaps only with the day-ahead cutover, not with the intraday one. Granular pricing only has economic content where within-hour residual demand actually varies, so we partition clock-hours into "critical" (the morning ramp and the evening peak, where the standard deviation of 15-minute residual demand is high) and "flat" (overnight, the within-day control).

Questions answered by this section.

1. How do day-ahead and intraday auctions relate, and where does regulated redispatch sit on top of them?
2. What changed mechanically on 11 December 2024, 19 March 2025 and 1 October 2025, and what did not?
3. Why is the critical/flat partition the right place to look for a granularity effect, and why is the partition itself identification-relevant?

2.1 Three reforms and a blackout (Dec 2024, Mar 2025, Apr 2025, Oct 2025)

The December 2024 settlement reform (ISP15) moves the imbalance settlement period from 60 to 15 minutes but leaves the wholesale contract grid unchanged. The intraday cutover (19 March 2025) moves all three intraday auctions and the continuous intraday market to 15-minute trading; the day-ahead cutover (1 October 2025) does the same for the day-ahead market. The 28 April 2025 blackout triggers a continuous reinforced-operation regime that overlaps only with the day-ahead cutover.

2.2 Where granular pricing has economic content: the critical/flat partition

A clock-hour is "critical" if the standard deviation of 15-minute residual demand exceeds the cross-hour median, and "flat" otherwise; the flat hours serve as the within-day control group for the bid-shape difference-in-differences. Critical here is the morning-ramp $\cup$ evening-ramp union; midday (solar-saturated) is excluded from the main specifications and reserved as a falsification.

3 Data: OMIE Bid-Level Offers, ENTSO-E Weather and Load, ESIOS Redispatch

We use OMIE per-curve sell offers for every unit, date and period in the day-ahead and intraday-auction markets from 2022 to 2026, aligned to the corresponding OMIE clearing prices and complemented by 15-minute ENTSO-E wind, solar and load and by ESIOS Fase I/II and aFRR regulated-redispatch volumes and prices. Bid tranches are restricted to a window-and-market-specific band around each clearing price (set to the $p_{90} - p_{50}$ spread of the MCP distribution in the analysis window) to isolate the bids that plausibly compete with the marginal price; the critical/flat partition is calibrated on post-15-minute-trading data.

Questions answered by this section.

1. What is a "curve" here, and what bid-shape measures are defined on it?

2. Why a window-specific bandwidth, and how sensitive are the conclusions to this choice?
3. How are OMIE's three offer types (simple, MIC, block) used in the analysis, and why does the offer-type composition matter for the bid-shape DiD?

3.1 Per-curve bid-shape outcomes and the in-band offered energy

Each (unit, date, period) bid ladder yields two per-curve bid-shape measures: the MW-weighted in-band standard deviation of bid prices σ_p and the effective tranche count N_{eff}. A complementary aggregate-volume primitive is the in-band offered energy E^{band} (in MWh per class-hour, normalised so that the quarter-count expansion under 15-minute trading does not mechanically inflate the level).

3.2 Calibrating the critical/flat partition (post-only)

Date-fixed-effect regressions of within-hour dispersion on critical and midday dummies, fit on post-15-minute-trading data, confirm that critical hours are where within-hour bid-shape dispersion actually lives across all three dispatchable techs.

4 Empirical Strategy: Three Specifications for Three Economic Objects

The reform's effect lives in three economic objects — bid shape, in-band offered energy, and auction-clearing outcomes — and each calls for a different specification. The bid-shape DiD identifies the critical-minus-flat shift across each cutover under within-day parallel trends, with unit fixed effects and date-clustered standard errors. The auction-clearing and aggregate-volume outcomes use a Bayesian counterfactual model (BSTS / CausalImpact) with wind, solar and gas covariates plus a 2024 same-calendar placebo, mirroring Märkle-Huß et al. (2017). Identification of the day-ahead cutover requires holding the post-blackout reinforced-operation regime constant: the day-ahead pre-window begins on 28 April 2025 (the day of the blackout) and ends on 30 September 2025, so both pre and post sit inside the *reforzada* period.

Questions answered by this section.

1. Why difference-in-differences for bid shape and a Bayesian counterfactual for prices and quantities?
2. What identifying assumption does the day-ahead pre-window choice rely on, and what would break with a naive long pre?
3. How does the 2024 same-calendar placebo correct for residual seasonal trend?

4.1 Spec A: Bayesian counterfactual for prices, cleared MW, and the day-ahead/intraday wedge SD

Daily outcomes (clearing prices, per-tech cleared GWh, within-day wedge SD) are modelled with a state-space local-linear trend, weekly seasonality and wind/solar/gas regressors trained pre-cutover and extrapolated forward; the post-cutover gap from the counterfactual is the treatment effect, with a 2024 same-calendar placebo subtracted for the joint 95% test.

4.2 Spec B: per-hour-class BSTS on in-band offered energy

The same Spec A architecture is run independently per (tech, market, hour-class) cell on E^{band}, the in-band offered energy normalised to MWh per class-hour. The readable summary is the real posterior and the 2024 placebo posterior reported side by side per hour-class; the reform-attributable shift is the gap.

4.3 Spec C: per-curve DiD for bid shape

Per-curve DiD on σ_p and N_{eff} with unit fixed effects and the critical/flat contrast. The DiD is the right tool here precisely because the granularity reform's effect on bid shape is structurally a critical-hours effect: differentiated bidding only pays off where within-hour residual demand actually varies. The critical $-$ flat differential θ *is* the granularity-enabled discrimination, not a residual being netted out.

5 Results

The empirical findings sort into five clusters, layered from bidding behavior up to auction-clearing outcomes and cross-market spillovers. *Bid-shape widening* shows up under both cutovers and concentrates in critical hours where the steep residual-demand curve rewards within-quarter discrimination. *In-band offered energy* migrates toward the newly-granular market under both reforms, with the morning ramp leading under the day-ahead cutover. On the auction-clearing side, the intraday reform drops both intraday and day-ahead clearing prices through a cross-market anticipation channel; the day-ahead reform scales up combined-cycle gas auction-cleared volume in critical hours by ~ 59 GWh/day; the day-ahead/intraday wedge keeps its mean at zero but its within-day standard deviation jumps under the intraday cutover. Finally, a *cross-market substitution* pattern — combined-cycle gas widens its day-ahead bid-shape *in anticipation* of finer intraday re-trading under the intraday cutover, and tightens its intraday bid-shape under the day-ahead cutover — shows that the reform restructures the whole sequential-market architecture, not just the auction it directly touches. The *operational vs. strategic* distinction explains why wind shows null DiD: wind aggregators face operational (imbalance-driven) incentives that apply equally in every hour and so leave no critical-flat differential, while dispatchable firms face strategic (rent-maximising) incentives that concentrate in critical hours.

Questions answered by this section.

1. Which findings survive both a within-pre placebo and a 2024 same-calendar placebo, and which need the joint 95% bracket?
2. Why does the bid widening translate into more day-ahead cleared volume but not into a day-ahead wholesale price change?
3. How does the wedge-SD jump reconcile with the wedge-level null — and what does that say about the within-hour restructuring?

5.1 Bid-shape widens in critical hours — both reforms (Spec C, Finding 1)

Combined-cycle gas, hydro and pumped-hydro significantly widen their within-quarter bid-price spread σ_p and effective tranche count N_{eff} in critical hours after each reform; flat hours barely move. Wind widens with magnitudes an order of magnitude smaller — the operational vs. strategic distinction is what collapses its critical-flat differential, not pure price-taking.

5.2 In-band offered energy migrates into the granular market — both reforms (Spec B, Finding 2)

The dispatchable bid-side migration shows up directly as MWh shifting in-band toward the newly-granular market: under the day-ahead reform the MWh move into the day-ahead auctions (morning ramp dominates), under the intraday reform into the intraday auctions.

5.3 Auction-clearing outcomes (Spec A, Finding 3)

The intraday reform drops both intraday and day-ahead clearing prices by $\sim$ 80 EUR/MWh placebo-net (the day-ahead drop is the rational-anticipation channel into day-ahead); the day-ahead reform scales up combined-cycle gas day-ahead cleared volume in critical hours by $\sim$ 59 GWh/day; renewables reallocate cleared MW toward the granular market under both reforms (ISP15-driven imbalance avoidance); the intraday reform leaves the per-clock-hour day-ahead $-$ intraday wedge level at zero but jumps its within-day SD in critical hours.

5.4 Cross-market substitution: the reform restructures the whole sequential-market architecture (Spec C, Finding 4)

Under the intraday reform, combined-cycle gas widens its day-ahead σ_p *in anticipation* of finer intraday re-trading; under the day-ahead reform, combined-cycle gas tightens its intraday σ_p, pulling structured bidding into the now-granular day-ahead. The granularity instrument rearranges the whole sequential market, not just the auction it directly touches.

5.5 Operational vs. strategic incentives explain the wind/solar null DiD (Finding 5)

Wind and solar aggregators bid at quarter granularity in every hour to manage imbalance exposure under the December 2024 settlement reform; that engagement is real but uniform across hour-classes, so the critical-flat DiD collapses for the renewables even though they are not pure price-takers. Dispatchable firms have no comparable operational pressure and engage strategically only where within-hour residual demand actually varies.

6 Robustness

The bid-shape and auction-clearing findings survive the standard checks: a within-pre midpoint placebo (the intraday-reform bid-shape findings are clean; the day-ahead-reform combined-cycle gas σ_p widening cleans up after de-trending), a midday-vs-flat hour-class falsification (sign-stable, smaller magnitudes), a cross-border-flow control on the intraday σ_p (essentially unchanged), a per-CCGT-firm decomposition (the widening is driven by two of the four firms, the two most active in the day-ahead competing tier), a bandwidth sensitivity check (the p_{90} choice isolates the competing tier from the parked scarcity tier; wider bandwidths contaminate the measure), a per-IDA-session

BSTS for the intraday price drop (session-symmetric), and an *offer-type composition check*: the combined-cycle gas in-band offer mix does shift across the day-ahead reform (simple-offer share drops 29% → 22%, block share grows 30% → 35%), but in the direction that should dampen the pooled σ_p widening; restricting to simple offers only strengthens the effect from 1.25^{**} to 3.53^{***}, so the pooled headline is a conservative read of the true within-tranche strategic engagement.

Questions answered by this section.

1. Which findings rely on a single robustness leg, and which on multiple?
2. Why is the offer-type composition shift not a confounder for the bid-shape DiD?
3. Why is *reforzada* a level confound rather than a within-hour confound?

7 Mechanism: Where the Strategic Margin Lives and Where Granularity Efficiency Shows Up

Two market-structure facts frame the bidding evidence. First, each Big-4 combined-cycle fleet bids a two-tier day-ahead curve — a "competing" tier near marginal cost and a "parked" tier above the clearing-price ceiling — and a single competitive tier at marginal cost in the intraday auction, indicating that the strategic margin is collected through regulated-redispatch recall, not through wholesale price-setting. Second, wholesale concentration (HHI $\sim$ 1,100) is moderate while ancillary-market concentration (HHI $>$ 4,000) is high, so market power migrated from wholesale to ancillary services as renewables eroded the wholesale margin. On the cost side, the granularity reforms compress the up-direction adjustment-cost channels REE has to call after the auction (real-time redispatch and aFRR reserve activations both shrink across the reform sequence) — the efficiency channel of 15-minute trading. The post-blackout reforzada cost rise concentrates on combined-cycle gas (Fase I up balloons; other technologies are flat) and is a separate policy effect that happens to overlap calendar-wise with the day-ahead cutover but operates on a different mechanism.

Questions answered by this section.

1. What is the geometry of the two-tier combined-cycle gas day-ahead curve, and why is the parked tier never recalled by the wholesale auction?
2. Why is the absence of wholesale price effects under the day-ahead reform consistent with the cleared-volume scale-up, the cross-market substitution, and the migration of strategic margin to ancillary services?
3. How do the granularity efficiency gains and the reforzada-driven cost rise interact in calendar time, and how do we keep them separate in the analysis?

7.1 Two-tier combined-cycle gas day-ahead curve and the day-ahead/intraday contrast

The same fleets bid a single competitive tier with negligible withholding in the intraday auction, after Fase I has cleared — indicating that the parked tier is a non-strategic placeholder rather than a Fase I bid, and the strategic margin is collected through a separate restriction offer submitted to REE.

7.2 Concentration migrated from wholesale to ancillary services

Wholesale-market HHI hovers around 1,100 over 2016–2024 and falls mid-day when solar peaks; ancillary-market HHI sits above 4,000 across the same window. The cleared-volume scale-up under the day-ahead reform is the combined-cycle gas competing tier expanding into the four quarters of each peak hour, scaling up the volume base that feeds the redispatch recall.

7.3 Granularity compresses the adjustment-cost channels — the efficiency channel

Finer trading-period resolution lets the auction itself absorb within-hour variation that previously had to be cleaned up by ex-post REE intervention. Real-time redispatch falls roughly threefold across the reform sequence; aFRR follows. Solar aggregators exploit granularity in midday hours by committing to four quarter-level schedules that track cloud-cover variation, instead of being locked into a single hourly commitment.

7.4 Reforzada cost increase concentrates on combined-cycle gas — a separate mechanism

Decomposing weekly dispatch into its components, the combined-cycle gas day-ahead-cleared layer collapses post-blackout while the intraday + REE real-time top-up layer balloons; other technologies are flat. A granularity reform that touched all bid-side firms equally would not produce a combined-cycle-gas-only intervention pattern — the reforzada cost rise is a separate policy effect from the granularity reforms.

8 Conclusion: Granularity Is a Strategic-Latitude Reveal

A market-microstructure reform that simply increases trading-period resolution acts not as a uniform technology shock but as a *strategic-latitude reveal*: it does not change marginal costs, capacity, or who is in the market; it changes who has room to discriminate within the hour and where, and the equilibrium response traces exactly that — strategic firms moving in strategic hours, operational firms moving uniformly, both rippling across the sequential markets through cross-market anticipation, and all of it within (not against) the competitive clearing. The reform does not amplify market power (wholesale prices fall, the wedge level is unchanged) and does not harm consumers (the granularity reforms compress the real-time adjustment-cost channels). The Fase I cost rise on the same calendar window is the post-blackout reinforced-operation response, not the granularity reforms: the two policies overlap in time but operate on different mechanisms. Whether the March 2025 intraday reform was in some way responsible for the April 2025 blackout is out of scope here — the ENTSO-E Expert Panel's final report does not list the reform as a contributing factor; the CNMC report does flag the move to quarter-hourly trading as one contributor to the abrupt voltage variations observed in the months around the incident. Answering it is beyond the scope of this thesis (it requires electrical-engineering knowledge we do not have) but it points to a strong research gap: the ENTSO-E report identifies the proximate technical triggers but does not engage with whether the market-microstructure reforms contributed to the macro-conditions that made the system fragile enough for those triggers to cascade.

Questions answered by the conclusion.

1. What is the main empirical lesson, and what is the surviving level-shift finding?
2. What is the interpretive frame, and what would tighten it from descriptive to identified?
3. What extensions follow naturally (the ongoing functional-factor regression for the full curve, a firm-level decomposition, identifying the redispatch channel)?